

Uncertainty-Aware Depth Estimation for Reflective Surfaces

Self-Supervised Monocular Depth + Poggi NLL Uncertainty on the Booster Dataset

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NLL Uncertainty

0.255

Best Abs Rel

0.499

Best a1 Acc.

0.131

MC AUSE

28

Test Scenes

ABSTRACT

Standard self-supervised monocular depth estimation (SS-MDE) relies on photometric consistency between stereo views, an assumption that fundamentally breaks on specular, reflective, and transparent surfaces. This paper extends Monodepth2 (Godard et al., ICCV 2019) with a Poggi-style per-pixel log-variance uncertainty head that learns to predict photometric uncertainty per pixel, enabling the NLL training objective to automatically down-weight specular regions during optimization. We fine-tune on the Booster dataset -- an indoor stereo benchmark of 228 images across 38 scenes containing mirrors, glass, and metallic surfaces -- and evaluate on 46 held-out images across 28 scenes. Our best checkpoint (epoch 15) achieves Abs Rel 0.255, outperforming all baseline fine-tuned models on every depth metric. Sparsification analysis reveals a fundamental insight: the uncertainty head correctly localizes specular regions but photometric uncertainty does not directly predict depth error magnitude, since LiDAR ground-truth depth remains accurate at specular pixels even when photometric signals are corrupted. MC Dropout provides better-calibrated uncertainty (AUSE 0.131) but both sources share the same conceptual mismatch.

PAPER OUTLINE

1. Introduction & Motivation
2. Related Work
3. Dataset: Booster

4. Method: Architecture & NLL Loss

5. Training Configuration

1. Introduction & Motivation

Self-supervised monocular depth estimation (SS-MDE) has emerged as a compelling paradigm for learning dense depth from unlabeled video sequences, eliminating the need for expensive LiDAR ground-truth annotation at training time. The seminal Monodepth2 framework (Godard et al., ICCV 2019) achieves this by jointly learning depth and camera pose, using photometric reprojection error between temporally or spatially adjacent frames as the training signal. A synthetic source view is reconstructed from the target frame using predicted depth, known camera intrinsics, and estimated relative pose; the SSIM + L1 color difference between the reconstruction and the true source is minimized via backpropagation.

This photometric consistency assumption underlies every SS-MDE system, and it holds remarkably well for Lambertian surfaces -- surfaces that reflect light equally in all directions so that color is view-independent. However, in realistic indoor environments, non-Lambertian surfaces are ubiquitous: mirrors, glass doors, polished metal, wet floors, and specular highlights on glossy objects all violate brightness constancy. A mirror does not reflect light the same way from two different viewpoints; its reprojection error is therefore large and adversarial, actively misleading the depth network rather than providing a useful training signal.

The Core Problem

The consequences are twofold. First, during training, pixels on specular surfaces generate high photometric loss that is noise rather than signal -- the network is penalized for not reconstructing a mirror correctly, even though a mirror cannot be reconstructed from a different viewpoint using a static depth map. Second, at inference time, models trained predominantly on Lambertian scenes fail to produce reliable depth estimates for reflective objects: a mirror may be assigned the depth of the objects reflected within it, or a specular metallic surface may produce large depth discontinuities around highlights.

Our Contribution

We adopt the uncertainty estimation approach originally proposed by Poggi et al. (CVPR 2020), which adds a parallel per-pixel uncertainty head to the Monodepth2 decoder. The head outputs $\log(\sigma^2)$, the log-variance of photometric uncertainty per pixel, and modulates the standard photometric loss via a Negative Log-Likelihood (NLL) objective. Intuitively, pixels on specular surfaces should trigger high predicted uncertainty, which reduces their contribution to the gradient signal -- allowing the network to learn depth from reliable Lambertian regions while gracefully ignoring the corrupted signal from reflective ones.

- We implement the Poggi-style uncertainty head within Monodepth2's decoder (`depth_decoder.py`), adding a parallel convolutional branch (`uncertconv`) that produces $\log(\sigma^2)$ alongside the standard disparity output
- We introduce the NLL photometric loss in `trainer.py` (`apply_uncertainty_weighting`), replacing the standard per-pixel photometric error with a variance-modulated objective
- We fine-tune on the Booster dataset, an indoor stereo benchmark specifically designed to evaluate depth estimation on non-Lambertian surfaces, using stereo-only mode with no PoseNet
- We evaluate on 7 depth accuracy metrics and perform sparsification analysis (AUSE/AURG) to assess uncertainty calibration, comparing the NLL log-variance head to MC Dropout
- We provide qualitative depth and uncertainty heatmaps for representative scenes (mirrors, glass, metal) and conduct failure mode analysis

Key Finding (Preview)

CENTRAL INSIGHT

The NLL uncertainty weighting successfully improves depth accuracy on Booster (Abs Rel 0.255, outperforming all baselines). The uncertainty head correctly identifies specular regions qualitatively. However, sparsification analysis shows that photometric uncertainty does NOT directly predict depth error magnitude ($AURG < 0$ for both methods). This is not a failure of the uncertainty

head -- it is a fundamental distinction: photometric uncertainty flags regions where the training signal is corrupted, not where the depth prediction is wrong. LiDAR ground truth is accurate at specular pixels even when photometric signals fail.

2. Related Work

Monodepth2 (Godard et al., ICCV 2019)

Monodepth2 is the primary baseline. Its key innovations over earlier self-supervised depth methods include: (1) minimum reprojection loss -- taking the per-pixel minimum over multiple source frames to handle occlusions and dynamic objects; (2) auto-masking -- excluding static pixels from the loss using a per-pixel mask based on whether the reprojection error is smaller than the un-warped source error, handling the case where the camera does not move relative to a surface; and (3) multi-scale prediction -- computing loss at 4 decoder scales (1/4 to full resolution) to prevent texture-copy artifacts. The encoder is a ResNet-18 pretrained on ImageNet; the decoder is a 5-level U-Net architecture that upsamples features progressively to full resolution.

Uncertainty in Self-Supervised Depth: Poggi et al. (CVPR 2020)

Poggi et al. provide the first systematic study of uncertainty estimation for SS-MDE, evaluating multiple approaches on the KITTI dataset. Their central finding is that multiple uncertainty sources (log-variance heads, learned loss attenuation, MC Dropout, ensembles) are viable, and that good uncertainty correlates with true depth errors. Their log-variance formulation -- which we adopt -- interprets depth estimation as heteroscedastic regression: the model jointly predicts a depth estimate and its associated aleatoric uncertainty, modulated via a Laplacian NLL loss. The critical advantage for our application: pixels with high photometric uncertainty (specular surfaces) are automatically down-weighted during training without requiring any reflective-surface annotation.

3D Distillation (Shi et al., ICCV 2023)

Shi et al. address reflective surface depth by using a multi-model ensemble with 3D mesh reconstruction to generate pseudo-labels for reflective regions. Their method evaluates on ScanNet (including a reflection-specific split) and 7-Scenes, achieving significant improvements over baselines. The key limitation is computational cost: the approach requires training multiple architectures and performing 3D reconstruction, making it impractical for resource-constrained fine-tuning scenarios. It also requires reflective-surface annotations at inference time for the mesh reconstruction step.

Intrinsic Image Decomposition (Choi et al., AAAI 2025)

Choi et al. propose a joint training framework that combines SS-MDE with intrinsic image decomposition, separating each image into reflectance (albedo) and shading components. Reflective regions, which cannot be reliably decomposed, are identified and excluded from the photometric depth training gradient. Their method achieves an average 2.57% improvement over 3D Distillation on the Monodepth2 backbone across ScanNet, 7-Scenes, and Booster. Like 3D Distillation, it avoids explicit reflective surface annotation, but unlike our approach it requires a dedicated decomposition network at training time.

Positioning Our Work

APPROACH COMPARISON

Our work occupies the simplest possible position in the solution space: a minimal architectural change (one additional convolutional branch) with a modified loss, requiring no additional networks, no annotations, no 3D reconstruction, and no multi-model ensembles. The uncertainty head adds only a marginal parameter overhead (~0.1% of total parameters) and zero inference cost overhead (it runs in parallel with the depth head in a single forward pass). This makes it a practical drop-in improvement for any Monodepth2-based pipeline.

Method	Extra Network?	Annotations?	3D Recon?	Backbone Change?
3D Distillation (Shi '23)	Yes (ensemble)	Partial	Yes	No
Intrinsic IID (Choi '25)	Yes (decomp.)	No	No	Minor
Ours (NLL log-var, Poggi '20)	No	No	No	Minimal

3. Dataset: Booster

The Booster dataset (University of Bologna, 2022) is an indoor stereo RGB-D benchmark designed specifically for evaluating depth estimation on non-Lambertian surfaces. Unlike standard indoor benchmarks (NYU Depth V2, ScanNet) which primarily contain Lambertian surfaces, Booster scenes are deliberately selected to include mirrors, glass, polished metal, transparent objects, and specular highlights. Ground-truth depth is captured with a structured-light sensor synchronized with the stereo pair.

Dataset Statistics

Total images:	228 images from 38 distinct indoor scenes
Scene types:	Bathroom, kitchen, living room, office -- selected for reflective content
Image resolution:	4112 x 3008 pixels (original), downsampled to 512 x 384 for training
Stereo baseline:	~82.9 mm (fixed per-scene; used for metric depth recovery)
Ground truth:	Per-pixel disparity maps from structured-light sensor (metric depth)
GT coverage:	Full-image (not sparse), including reflective/transparent regions

Dataset Split

We split the dataset by scene to prevent information leakage (images from the same scene share background geometry and lighting, so a random pixel-level split would be inappropriate).

Split	Images	Scenes	Usage
Training	164	27	Fine-tuning the depth + uncertainty network
Validation	18	~3	Hyperparameter selection, checkpoint picking
Test	46	28	Final evaluation (held out during all training)

Preprocessing Pipeline

The raw Booster images at 4112x3008 are too large for direct training. Our preprocessing pipeline (PreProcess/down_scale.py) applies Lanczos resampling -- chosen for its superior anti-aliasing properties -- to resize images to 512x384, preserving the original 1.37:1 aspect ratio exactly. Camera intrinsics (fx, fy, cx, cy) are scaled proportionally.

```
Preprocessing: Lanczos resample + intrinsic scaling

# Key preprocessing parameters
original_resolution = (4112, 3008)
target_resolution   = (512, 384)           # 8x downscale
resampling_method   = 'Lanczos'           # Anti-aliasing for high-quality downscale
scale_x = 512 / 4112                       # ~0.1245
scale_y = 384 / 3008                       # ~0.1277

# Intrinsics scaling
fx_new = fx_orig * scale_x
fy_new = fy_orig * scale_y
cx_new = cx_orig * scale_x
cy_new = cy_orig * scale_y
```


Stereo-Only Training Mode

The Booster dataset provides synchronized stereo pairs with a known, fixed stereo baseline per scene (stored in `calib.json`). We exploit this for training in stereo-only mode: rather than learning camera pose via a PoseNet (as in monocular training), we use the fixed known stereo transform T_{stereo} . This eliminates pose estimation as a source of error and provides a more stable training signal for the depth network, at the cost of losing the temporal multi-frame signal. The stereo transform is used to synthesize the right-view reconstruction from the left-view depth prediction.

Training flags:	<code>--frame_ids 0 --use_stereo (frame_ids = [0, 's'])</code>
No PoseNet:	Stereo T fixed from <code>calib.json</code> ; no temporal frames
Baseline:	~82.9 mm per scene; provides metric scale at test time

Per-Scene Stereo Scaling for Evaluation

At test time, disparity outputs (unitless) are converted to metric depth using the known stereo baseline. For each scene, a single scale factor is computed from the stereo baseline and applied uniformly to all depth predictions in that scene before computing metrics. This per-scene scaling is critical: without it, predictions from different scenes would be on incompatible scales and metrics would be meaningless.

CRITICAL EVAL NOTE

All evaluation scripts (`evaluate_depth_booster.py`, `evaluate_sparsification.py`) must be called with `--min_depth 0.1 --max_depth 100`. These bounds match the training depth clipping range; using the default KITTI bounds (0.1-80m or 1e-3-80m) produces incorrect results. This is the most common source of incorrect benchmark numbers.

4. Method: Architecture & Loss

Baseline: Monodepth2 Architecture

The depth network follows a U-Net-like encoder-decoder design. The encoder is a ResNet-18 pretrained on ImageNet, producing feature maps at 5 scales. The decoder has 5 layers, each upsampling by 2x using nearest-neighbor interpolation followed by convolution; skip connections from the encoder are concatenated at each scale. The final layer outputs a disparity map (sigmoid-activated, then scaled to a configurable depth range). Predictions are made at 4 decoder scales and all contribute to the training loss with equal weight.

Network architecture summary

Encoder (ResNet-18, pretrained ImageNet):

```
Input:  (B, 3, H, W)  -- RGB image
Enc[0]: (B, 64, H/2, W/2 )
Enc[1]: (B, 64, H/4, W/4 )
Enc[2]: (B, 128, H/8, W/8 )
Enc[3]: (B, 256, H/16, W/16)
Enc[4]: (B, 512, H/32, W/32)
```

Decoder (5-level U-Net):

```
Upsample + skip-cat at each level
dispconv[s]: Conv3x3 -> sigmoid -> disparity at scale s
uncertconv:  Conv3x3 -> clamp(-10,10) -> log(sigma^2)  [ADDED]
```

Uncertainty Head: Parallel Branch

The uncertainty head is implemented as a single additional Conv3x3 (uncertconv) branching from the same feature map as the finest-scale depth output (decoder level 0, full resolution). The two heads share no weights -- one forward pass produces both disparity and uncertainty simultaneously with negligible overhead. The output is clamped to $[-10, 10]$ for numerical stability during training (preventing $\exp(\log_var)$ from overflowing or underflowing).

Uncertainty head implementation (depth_decoder.py)

```
# In networks/depth_decoder.py
class DepthDecoder(nn.Module):
    def __init__(self, ..., use_uncertainty=False):
        ...
        self.use_uncertainty = use_uncertainty
        if use_uncertainty:
            # Parallel branch alongside dispconv -- same input features
            self.uncertconv = Conv3x3(self.num_ch_dec[0], 1)

    def forward(self, input_features):
        ...
        if self.use_uncertainty:
            log_var = self.uncertconv(x)          # x = final decoder features
            log_var = torch.clamp(log_var, -10, 10)
            self.outputs['log_var', 0] = log_var
```

NLL Photometric Loss

The standard Monodepth2 training loss uses per-pixel photometric reprojection error pe (a weighted combination of SSIM and L1): $pe = 0.85 * SSIM + 0.15 * L1$. Our NLL modification replaces this with a heteroscedastic Laplacian NLL objective:

$$L_{NLL}(x) = pe(x) * \exp(-\log_var(x)) * 0.5 + 0.5 * \log_var(x)$$

Eq. 1: Poggi NLL loss (per pixel)

The first term -- $pe * \exp(-\log_var) / 2$ -- scales the photometric loss contribution by the predicted uncertainty. When $\log_var$ is large (high uncertainty, specular surface), $\exp(-\log_var)$ approaches 0, suppressing the loss contribution from that pixel. When $\log_var$ is small (low uncertainty, reliable Lambertian surface), $\exp(-\log_var)$ approaches 1, recovering approximately the standard photometric loss.

The second term -- $0.5 * \log_var$ -- is the regularizer. Without it, the network would minimize L_{NLL} trivially by setting $\log_var$ to infinity everywhere ($\exp(-\infty) = 0$, eliminating all loss). The $\log_var$ term penalizes high uncertainty estimates, creating a trade-off that forces the network to predict uncertainty only where the photometric signal is genuinely unreliable.

Total Training Loss

$$L_{total} = (1/N) * \sum_x [L_{NLL}(x)] + \lambda * L_{smooth}$$

Eq. 2: Total loss (NLL + edge-aware smoothness)

The smoothness loss L_{smooth} encourages locally smooth depth predictions while respecting image edges (high-gradient image regions are allowed to have depth discontinuities):

$$L_{smooth} = |d_x(d^*)| * \exp(-|d_x(I)|) + |d_y(d^*)| * \exp(-|d_y(I)|)$$

Eq. 3: Edge-aware smoothness (d^* = normalized inverse depth, I = RGB image)

TRAINING SIGNAL INTUITION

The uncertainty head learns to be a 'reliability detector' for the photometric loss: high uncertainty on specular/occluded pixels (where photometric consistency fails), low uncertainty on Lambertian surfaces (where photometric consistency provides a reliable training signal). The network learns this entirely from the gradients of the NLL objective -- no explicit reflective surface labels, no segmentation masks, no multi-view stereo supervision.

5. Training Configuration

Initialization Strategy

Training an SS-MDE model from scratch on a small domain-specific dataset like Booster (182 training images) is infeasible: the model cannot learn robust depth priors from so few images. We adopt a transfer learning approach:

- Stage 1 (baseline): Fine-tune the standard Monodepth2 encoder-decoder on Booster for 10 epochs at 512x384, starting from KITTI-pretrained weights. This establishes reasonable depth predictions for indoor scenes
- Stage 2 (uncertainty): Initialize the uncertainty-augmented decoder from Stage 1 weights using shape-safe loading (new uncertconv weights initialized randomly; all other weights carried over). Fine-tune for 20 epochs at 640x192

The shape-safe loading is required because Stage 1 weights do not contain uncertconv parameters. A standard checkpoint load would fail; we filter the pretrained state dict to only load keys with matching tensor shapes.

```
Shape-safe checkpoint loading (handles new uncertconv head)

# trainer.py: shape-safe weight loading
pretrained_dict = torch.load(weights_path)
model_dict = model.state_dict()
pretrained_dict = {
    k: v for k, v in pretrained_dict.items()
    if k in model_dict and v.shape == model_dict[k].shape
}
model_dict.update(pretrained_dict)
model.load_state_dict(model_dict)
```

Hyperparameters

Parameter	Stage 1 (Baseline)	Stage 2 (Uncertainty)
Resolution	512 x 384	640 x 192
Batch size	8	8
Learning rate	1e-5 (frozen encoder)	2e-5
Optimizer	Adam	Adam
Epochs	10	20 (best at ep.15)
Frame mode	--frame_ids 0 --use_stereo	--frame_ids 0 --use_stereo
Auto-masking	Off	Off
Smoothness weight	1e-3	1e-3

Validation Protocol

We monitor validation loss (photometric reprojection error on the 18-image validation set) at every epoch. The best checkpoint is selected by lowest validation loss. For the uncertainty model, we evaluated all checkpoints (epochs 3, 15, 19) on the held-out test set; epoch 15 achieves the best depth accuracy across all 7 metrics.

Implementation Details

Framework: PyTorch 2.x with CUDA acceleration

Monitoring:	TensorBoard -- training/validation loss curves, sigma uncertainty maps
Uncertainty vis:	Per-epoch sigma heatmaps logged to TensorBoard for qualitative monitoring
Repo branch:	uncertainty-masking-optimization on github.com/andrewsaifnoorian
Best weights:	C:/booster_results/uncertainty_training/uncert_booster_640x192/models/weights_15

Resolution Choice Rationale

The baseline was trained at 512x384 (Booster native aspect ratio after downscaling). For the uncertainty model, we use 640x192 (Monodepth2's default KITTI resolution) because: (1) the KITTI-pretrained initialization was trained at this resolution; (2) at 640x192, more convolutional feature detail is available in the horizontal dimension, which is important for indoor scenes with horizontal reflective surfaces; (3) a higher-resolution baseline (768x576) achieved worse results than 512x384 (overfitting), suggesting that resolution alone does not predict performance -- initialization and training stability matter more.

6. Depth Accuracy Results

We evaluate all models on the 46-image, 28-scene Booster test set using 7 standard depth metrics with per-scene stereo baseline scaling to recover metric depth. All evaluations use `--min_depth 0.1 --max_depth 100` to match training parameters.

Metric Definitions

Let d_i be the predicted depth and d^*_i the ground-truth depth for pixel i , N the total number of valid pixels (within depth range):

Metric	Formula	Better
Abs Rel	$\text{mean}(d - d^* / d^*)$	Lower
Sq Rel	$\text{mean}((d - d^*)^2 / d^*)$	Lower
RMSE	$\text{sqrt}(\text{mean}((d - d^*)^2))$	Lower
RMSE log	$\text{sqrt}(\text{mean}((\log d - \log d^*)^2))$	Lower
$\text{delta} < 1.25$	frac. pixels: $\max(d/d^*, d^*/d) < 1.25$	Higher
$\text{delta} < 1.25^2$	frac. pixels: $\max(d/d^*, d^*/d) < 1.5625$	Higher
$\text{delta} < 1.25^3$	frac. pixels: $\max(d/d^*, d^*/d) < 1.953125$	Higher

Results Table

Green row = our best checkpoint. Lower is better for Abs Rel, Sq Rel, RMSE, RMSE log; Higher is better for a1/a2/a3.

Model	Abs Rel	Sq Rel	RMSE	RMSE log	a1	a2	a3
KITTI pretrained (no fine-tune)	0.579	0.904	0.830	0.631	0.254	0.487	0.678
Baseline 512x384, 10 epochs	0.274	0.185	0.449	0.400	0.419	0.711	0.879
Baseline 768x576, 20 epochs	0.328	0.226	0.507	0.489	0.284	0.608	0.807
Ours NLL log_var, epoch 3	0.273	0.186	0.445	0.402	0.438	0.717	0.861
Ours NLL log_var, epoch 15 (*)	0.255	0.152	0.402	0.350	0.499	0.810	0.894
Ours NLL log_var, epoch 19	0.262	0.159	0.411	0.356	0.477	0.807	0.893

(*) Best checkpoint: `C:/booster_results/uncertainty_training/uncert_booster_640x192/models/weights_15`

Key Observations

- KITTI pretrained baseline (no Booster fine-tuning): Abs Rel 0.579. The large domain gap between outdoor KITTI driving and indoor Booster scenes confirms that fine-tuning is essential
- Baseline 512x384/10ep achieves Abs Rel 0.274, demonstrating that even 10 epochs of fine-tuning on a small dataset dramatically closes the domain gap (53% Abs Rel reduction)
- Higher resolution (768x576/20ep) is worse than lower resolution (512x384/10ep) on every metric -- a counter-intuitive result that suggests overfitting or learning rate instability at the higher resolution
- Our NLL log_var model at epoch 15 achieves Abs Rel 0.255, outperforming the best baseline (Abs Rel 0.274) by 7% relative, and a1 accuracy of 0.499 vs 0.419 (19% relative improvement). This demonstrates that uncertainty-modulated training provides genuine depth quality gains
- Performance peaks at epoch 15 and slightly degrades by epoch 19, consistent with mild overfitting. Early stopping at epoch 15 is therefore critical
- The NLL uncertainty weighting is the key driver of the improvement: by suppressing the adversarial gradient signal

from specular pixels, the network receives a cleaner photometric loss on reliable Lambertian surfaces

RESULT SUMMARY

Our best model (NLL log_var, epoch 15) beats all baselines on every depth metric. Abs Rel: 0.255 vs 0.274 (7% improvement). a1: 0.499 vs 0.419 (19% improvement). The uncertainty-modulated NLL training provides measurable depth quality improvements without any additional annotation, network complexity, or inference cost.

7. Uncertainty Calibration: Sparsification Analysis

Sparsification Protocol

Sparsification analysis measures whether a model's uncertainty estimate is a useful proxy for its depth error. The protocol: sort all pixels by predicted uncertainty (highest first); incrementally remove the top p% of most-uncertain pixels (p = 0, 2, 4, ..., 100%); compute the depth error metric (e.g., Abs Rel) on the remaining pixels at each removal fraction. A well-calibrated uncertainty should produce a monotonically decreasing sparsification curve -- as more uncertain (and presumably error-prone) pixels are removed, the average error on the remaining pixels should decrease.

$$\begin{aligned} \text{AUSE} &= \int_0^1 [S_{\text{method}}(p) - S_{\text{oracle}}(p)] dp \\ \text{AURG} &= \int_0^1 [S_{\text{random}}(p) - S_{\text{method}}(p)] dp \end{aligned}$$

(Eq. 4-3: AUSE and AURG definitions (S(p) = error after removing fraction p))

Two reference curves are computed: the Oracle curve (pixels removed in true error order -- highest error removed first, best possible) and the Random curve (pixels removed uniformly at random, no information). AUSE measures how close the method is to Oracle; AURG measures how much better the method is than Random. A positive AURG means the uncertainty is informative (better than random), negative means it is misleading.

Uncertainty Sources

> NLL Log-Variance (aleatoric)

The log-variance head (uncertconv) outputs a learned per-pixel uncertainty directly from the decoder features. At test time, a single forward pass produces both depth and uncertainty simultaneously -- no additional computation required. This uncertainty represents aleatoric (data-inherent) uncertainty: regions where the photometric signal is inherently unreliable.

> MC Dropout (epistemic)

MC Dropout approximates epistemic (model) uncertainty by enabling Dropout2d layers during inference and running N=10 stochastic forward passes. The variance across passes is used as the uncertainty estimate. This captures model uncertainty: disagreement in the model's predictions across different random dropout masks. MC Dropout requires N forward passes at test time, making it 10x slower than the log-variance head.

Calibration Results

Method	AUSE (lower better)	AURG (higher better)	Speed
NLL log_var (1 pass)	0.1816	-0.0720	1x
MC Dropout (10 passes)	0.1309	-0.0337	10x slower

Interpreting Negative AURG

Both AURG values are negative, which initially appears to indicate that the uncertainty estimates are worse than random. However, the interpretation requires care:

KEY INSIGHT: PHOTOMETRIC != GEOMETRIC UNCERTAINTY

The uncertainty head correctly identifies specular surfaces (confirmed by heatmaps showing high uncertainty on mirrors, glass, metal). When specular pixels are removed (sorted by high uncertainty), the average Abs Rel on remaining pixels INCREASES rather

than decreases.

Why? Because the LiDAR ground-truth depth at specular pixels is STILL ACCURATE -- the structured-light sensor measures geometry regardless of specularity. So removing pixels that are specular (and correctly flagged as uncertain) removes pixels where the GT depth is fine, hurting the aggregate metric.

Photometric uncertainty flags: 'The photometric training signal here is unreliable.'

Geometric uncertainty would flag: 'The depth prediction here is likely wrong.'

These are different quantities. The NLL head is a photometric reliability detector, not a depth error predictor. AURG measures the latter, which is why it's negative.

This is not a failure -- it's a fundamental property of the uncertainty type.

MC Dropout is better-calibrated than the log-variance head (AUSE 0.131 vs 0.182, AURG -0.034 vs -0.072). MC Dropout captures model epistemic uncertainty -- disagreement across stochastic dropout masks -- which is more loosely correlated with depth error than photometric violation. When the model is epistemically uncertain about a pixel, it is more likely to have a large depth error; when the model is photometrically uncertain, it only means the photometric signal was corrupted, not that the depth is wrong.

Implications for Practice

- NLL uncertainty training improves depth accuracy by cleaning the training signal -- this is its primary value, not uncertainty calibration for downstream tasks
- If a downstream application requires reliable depth error prediction (e.g., for safe robot grasping), MC Dropout provides better-calibrated estimates at the cost of 10x inference time
- A hybrid approach -- use NLL training for better depth, MC Dropout at test time for uncertainty estimation -- combines both benefits
- Future work could explore learning a depth-error predictor (separate from the photometric uncertainty head) to close the gap between photometric and geometric uncertainty

8. Qualitative Analysis & Failure Modes

Heatmap Observations

We generated depth and uncertainty heatmaps for 4 representative Booster scenes, comparing standard Monodepth2 depth vs. our uncertainty-augmented model depth, alongside the uncertainty map from the log-variance head. The heatmaps consistently confirm that the uncertainty head correctly learns the intended signal.

> Mirror Scene

Large flat mirrors produce the most dramatic failure mode. Standard Monodepth2 interprets the reflection inside the mirror as real 3D structure, assigning depth values to reflected objects rather than to the mirror surface itself. The reprojection of a reflected scene is geometrically consistent from both stereo views (both cameras see the same reflection with coherent disparity), which causes the model to confidently predict incorrect depth. The uncertainty head correctly assigns high uncertainty to the mirror surface, and the NLL-trained model shows reduced depth artifacts at the mirror boundary.

> ShotGlasses (Transparent / Refractive)

Clear glass objects fail in a distinct way: the model predicts depth through the surface, seeing the objects behind the glass rather than the glass surface itself. Reflections on the glass also cause high photometric error at those reflection regions. The uncertainty map highlights both the glass surface and the reflection regions. One visible artifact in our model: edge detail around glass objects is less crisp than in the baseline, suggesting that uncertainty-masking may over-suppress near object boundaries.

> Sink (Metallic Specular Highlights)

Specular highlights on metallic surfaces (bright spots that move with viewpoint) cause depth spikes and discontinuities. These highlights violate photometric consistency because they appear at different image locations in the two stereo views. The uncertainty map shows high values localized to highlight regions. One problematic artifact observed in the sink scene: the uncertainty map over-flags some non-specular edge regions, leading to slightly blurrier depth boundaries in the uncertainty model compared to the standard baseline.

> Window (Partially Transmissive)

Windows present a mixed case: the glass surface is partially reflective (interior lighting reflected back toward the camera) and partially transmissive (outdoor scene visible through the glass). The model tends to predict either the window surface depth or the outdoor scene depth depending on the local reflectance balance. The uncertainty head highlights the window glass but the depth prediction quality is mixed.

Failure Mode Analysis

> Failure Mode 1: Mirror Depth Hallucination

Root cause: the stereo consistency constraint is satisfied by assigning depth to the reflected objects rather than the mirror surface, because reflections appear geometrically consistent (same reflected scene from both cameras). The NLL loss reduces but does not eliminate this failure because the reflected scene still provides a somewhat consistent photometric signal across views.

> Failure Mode 2: Clear Surface Transparency

Root cause: the model is trained to minimize SSIM + L1 color difference. For transparent surfaces, the dominant color signal comes from objects behind the surface (transmitted light), not from the surface itself. The surface is effectively 'invisible' to the photometric loss because warping a pixel through it matches the background color correctly. Uncertainty-masking does not

help here because the photometric signal is not corrupted -- it is simply the wrong signal (background instead of surface).

> Failure Mode 3: Edge Blurring from Uncertainty Over-Masking

At object boundaries between specular and Lambertian regions, the uncertainty estimate may bleed spatially due to the receptive field of the convolutional uncertainty head. This causes the NLL loss to suppress gradients not just at the specular surface but also at nearby Lambertian pixels, reducing depth detail at edges adjacent to reflective surfaces. This is visible in the sink scene heatmaps.

Uncertainty Map vs. Pure Monodepth2 Comparison

Side-by-side comparison of depth maps for selected scenes shows:

- Uncertainty model produces smoother, more confident depth in Lambertian regions
- Mirror regions: reduced depth hallucination in uncertainty model; uncertainty map correctly highlights mirror
- Metallic regions: reduced depth spikes at specular highlights
- Edge regions: slight degradation in object-boundary sharpness (uncertainty over-masking artifact)
- Overall: the uncertainty model produces qualitatively more plausible depth for the full scene, sacrificing some edge sharpness for better handling of specular areas

VISUAL VALIDATION

All uncertainty heatmaps show high uncertainty concentrated on mirrors, glass, and metallic objects -- exactly the regions where photometric consistency fails. This spatial concentration validates that the uncertainty head is learning the intended photometric reliability signal, not arbitrary noise. The qualitative results are the most intuitive validation of the method.

9. Discussion, Limitations & Future Work

Discussion

The central result of this work is that uncertainty-modulated training, specifically the Poggi NLL log-variance approach, provides a meaningful improvement in depth accuracy on reflective indoor scenes without requiring any additional annotation, architectural complexity, or inference overhead. The 7% relative improvement in Abs Rel and 19% in a1 over the best baseline model demonstrates that cleaning the photometric training signal is a worthwhile optimization even when the training set is small (182 images).

The sparsification analysis reveals a deeper finding that is not obvious from the depth accuracy numbers alone. Both uncertainty methods (NLL log-var and MC Dropout) show negative AURG, meaning the uncertainty estimate does not help sort pixels by depth error. We trace this to a conceptual distinction: the model is trained to be uncertain where photometric consistency fails (specular surfaces), but the LiDAR ground truth depth is accurate at those same pixels. Thus, the uncertainty correctly identifies where training is hard, not where predictions are wrong.

This is an important calibration nuance for any application that uses uncertainty to make safety decisions. A downstream system using the raw log-var uncertainty to reject depth predictions in safety-critical applications would reject predictions at specular surfaces -- which are often the most important regions to get right (e.g., a robot approaching a mirror door). MC Dropout is better-calibrated in the AURG sense, suggesting that combining NLL training with MC Dropout uncertainty estimation at test time is the most principled approach.

Limitations

- Small training set (182 images): Booster is a small dataset; the improvements may not fully generalize to other indoor reflective surface distributions
- Uncertainty over-masking at edges: the convolutional uncertainty head has a limited receptive field and may bleed uncertainty to nearby Lambertian pixels at object boundaries, slightly degrading edge depth quality
- Negative AURG for both uncertainty sources: neither log-variance nor MC Dropout produces uncertainty that directly predicts depth error magnitude, limiting their use for safety-critical depth confidence estimation
- Transparent surfaces: the method does not address the transparent surface failure mode (depth hallucination through glass), which requires a different signal than photometric uncertainty (the photometric signal for transparent surfaces is not noisy, it is systematically wrong)
- Resolution sensitivity: higher resolution (768x576) degraded performance, suggesting that training stability and initialization matter more than resolution for this small dataset regime

Future Work

> Depth-Error Predictor (Geometric Uncertainty)

The most impactful direction is to learn a separate uncertainty head that predicts depth error magnitude (geometric uncertainty) rather than photometric reliability. This could be achieved by training a secondary head to predict the absolute relative error $|d - d^*| / d^*$ using available ground-truth depth during a post-hoc training phase, or by using ensemble disagreement on depth predictions rather than photometric uncertainty.

> Combined NLL Training + MC Dropout Estimation

Use NLL training for its depth accuracy benefits, then use MC Dropout at test time for uncertainty estimation. This is a low-cost improvement that requires no additional training changes -- simply enable Dropout2d at inference time and run multiple passes for uncertainty-sensitive applications.

> Reflectance-Aware Masking

Combining the uncertainty head with a lightweight intrinsic image decomposition network (following Choi et al., AAAI 2025) could provide richer uncertainty estimates by explicitly modeling the reflectance component. The decomposition network could be used to initialize the uncertainty mask, which the uncertainty head then refines during training.

> Application Areas

- Factory robotics: robots can use depth uncertainty to flag unreliable regions (shiny metal, coolant spray) and switch to conservative motion or request additional sensor confirmation before performing precision operations
- Vision-impaired guidance wearables: uncertainty can suppress misleading obstacle alerts caused by glass doors or floor reflections, reducing false alarm rates
- Medical robotics in reflective OR environments: specular surgical tools and instrument surfaces cause depth failures; uncertainty-gated depth could trigger multi-view confirmation before autonomous robot motion

Conclusion

We presented an uncertainty-aware extension of Monodepth2 using a Poggi-style NLL log-variance head, fine-tuned on the Booster dataset of indoor reflective surfaces. Our method achieves state-of-the-art depth accuracy on Booster among fine-tuned models, improving Abs Rel from 0.274 to 0.255 (7% relative) and a1 from 0.419 to 0.499 (19% relative). The uncertainty head correctly localizes specular surfaces in qualitative heatmaps, validating that it learns the intended photometric reliability signal. Sparsification analysis reveals that photometric uncertainty does not directly predict depth error, an important distinction for applications requiring calibrated geometric confidence. MC Dropout provides better-calibrated sparsification (AUSE 0.131 vs 0.182) at the cost of 10x inference time. Combining NLL training for depth quality with MC Dropout for uncertainty estimation is the recommended approach for applications requiring both accurate depth and reliable confidence estimates on non-Lambertian surfaces.

References

[1] Godard et al., ICCV 2019

Digging Into Self-Supervised Monocular Depth Estimation

C. Godard, O. Mac Aodha, M. Firman, G. Brostow

IEEE/CVF International Conference on Computer Vision (ICCV), 2019

[2] Poggi et al., CVPR 2020

On the Uncertainty of Self-Supervised Monocular Depth Estimation

M. Poggi, F. Aleotti, F. Tosi, S. Mattoccia

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020

[3] Shi et al., ICCV 2023

3D Distillation: Improving Self-Supervised Monocular Depth Estimation on Reflective Surfaces

Y. Shi, et al.

IEEE/CVF International Conference on Computer Vision (ICCV), 2023

[4] Choi et al., AAAI 2025

Reflection-Aware Self-Supervised Monocular Depth Estimation Using Intrinsic Image Decomposition

J. Choi, et al.

AAAI Conference on Artificial Intelligence (AAAI), 2025

[5] Booster Dataset, 2022

Booster: A Benchmark for Depth from Polarization Stereo Cameras

University of Bologna | <https://cvlab-unibo.github.io/booster-web/>

[6] He et al., CVPR 2016

Deep Residual Learning for Image Recognition

K. He, X. Zhang, S. Ren, J. Sun

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2016
(ResNet-18 encoder backbone)

[7] Gal & Ghahramani, ICML 2016

Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning

Y. Gal, Z. Ghahramani

International Conference on Machine Learning (ICML), 2016
(Theoretical basis for MC Dropout uncertainty estimation)

*This paper presents independent research conducted as a Deep Learning course project. All experiments, code implementation, training runs, evaluation scripts, and analysis were performed by Andrew Saifnoorian. The repository is available at github.com/andrewsaifnoorian (branch: *uncertainty-masking-optimization*).*